A Local Community Detection Method Based on Folded Subgraph

Mengting Zhang and Weihong Bi

***Abstract*—Community structure refers to the "small groups" in the network. Detecting community structure in networks has significant application value. With the continuous expansion and complexity of the network, the global information of the network is often difficult to obtain. On the other hand, in some cases, we pay more attention to the local community where the given node is located. Local community detection methods detect local community structure by using local information from a given node. However, many local community detection methods encounter the problem of precision limitation. Therefore, in order to alleviate such problems, we propose the FG-based method in this paper. Based on the characteristics of complex networks, a folded subgraph method is designed to consider some similar nodes as single nodes, reducing the impact of noise in the network. Furthermore, based on the folded subgraph, FG-based method designs a three-stage local expansion strategy, in which nodes with different characteristics are added to the local community in each stage. We conduct experiments on datasets and find that the FG-based method can improve the recall and precision of local community structures.**

***Index Terms*—Local community detection, precision limitation, local modularity, social network.**

I. Introduction

MANY systems can now be represented as complex networks [1], [2], such as social networks and protein-protein interaction networks. In recent years, numerous researchers have focused on analyzing complex networks to extract valuable information applicable across various disciplines. Community structure is one of the most crucial aspects of network analysis. From a qualitative standpoint, the community structure reveals that the nodes within the same community are densely interconnected, whilst the connections between communities are scant. Newman et al. [3], [4] developed the definition of modularity Q to quantitatively assess the quality of community structure. Community detection is the process of using topology information to determine the network's community structure. Many innovative methods, such as random walk [5], [6], label propagation [7], [8] and other typical methods [9], [10] have been presented for detecting communities. Community structure is widely used in real life, such as applied to simulated earthquake ground motion data in the Los Angeles region [12].

Received 12 February 2024; revised 16 March 2025; accepted 5 April 2025. Date of publication 21 April 2025; date of current version 28 May 2025. Recommended for acceptance by R. Akbarinia. *(Corresponding author: Weihong Bi.)*

The authors are with the School of Mathematical Sciences, Zhejiang University, Hangzhou 310000, China (e-mail: bivh@zju.edu.cn).

Digital Object Identifier 10.1109/TKDE.2025.3563100

With the rapid expansion of the Internet, numerous real-world networks have exhibited features of enormous scale. Deep learning techniques show an increasing capacity to encode feature representations of high-dimensional data. Thus, deep neural networks, deep graph embedding and graph neural networks are widely used in the field of community detection research [13]. In traditional community detection, global network information is often required. However, obtaining global information for large-scale networks is typically challenging. In addition, it frequently takes a considerable amount of effort to identify all community structures in vast networks. Therefore, most traditional community detection methods have limited applications. In such situations, numerous researchers concentrate on local community detection. Local community detection entails identifying the local community to which a given beginning node belongs based on local information. A number of efficient local community detection methods have also evolved in recent years. Clauset et al. introduced the definition of local modularity R in paper [14] and utilized the concept of greedy approach to maximize local modularity R as the clustering rule. Luo et al. defined a local modularity LQ in their article [15].

Since only local topology information is utilized in the calculation process, the time complexity of local community detection methods is typically low [16]. In addition, mining the local community structure of a node has significant application value. However, these methods also have some limitations, which are manifested in two ways. (1) To determine whether a node should join the local community, the method must employ the local modularity function.Such methods will be subject to the characteristics of modularity, leading to the problem of precision limitation that renders the detection of a single local community unacceptable in terms of recall and precision. (2) Due to the lack of neighborhood information and the influence of noise in the network, the results are often unsatisfactory. Thus, the local community detection method must develop appropriate expansion strategies. Many methods [17], [18] have also emerged to improve the above issues.

In this paper, we propose a new local community detection method FG-based. The main contributions of our research are as follows:

1) We propose a method known as Folded Subgraph, which leverages the relationship features of network nodes. Closely related nodes are compressed into a single node for subsequent expansion. On the one hand, the compressed node can provide more information to improve the precision of expansion. On the other hand, nodes with

1041-4347 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.

relatively low degrees can join the local community through their connections to other nodes, thereby improving recall.

2) Based on the Folded Subgraph method, we propose a method called FG-based which can be used for local community detection. In FG-based method, combined with the Folded Subgraph method, we further design a three-stage expansion strategy, in which each stage lets different types of nodes join the local community.
3) We test the method on several synthetic and real datasets. The experimental results demonstrate that our proposed method, FG-based method, significantly enhances the quality of the local community structure, i.e., the recall rate and precision rate. The FG-based strategy can mitigate the problem of precision limitation to some extent.

The rest of the paper is organized as follows. In Section II, related works on local community detection and precision limitation are introduced. We study real-world networks, observe that the community membership of nodes exhibits the following properties, and propose the Folded Subgraph method in Section III. In Section IV, we describe the three-stage expansion strategy of the FG-based method, and in Section V, we present and analyze experimental results. The selection of hyperparameters in the proposed method and the need for the three-stage extension are discussed in Section VI. Conclusions are presented in Section VII.

II. Related Work

A. Resolution Limitation

In 2002, Newman et al. [19] proposed the GN method, which sparked significant research interest in the field of community detection. Later, Newman et al. [4] proposed modularity Q as a metric to evaluate the quality of community detection results.

As an important evaluation metric of community detection, modularity Q is often used as the objective function of optimization methods [3]. Therefore, numerous classical optimization-based methods [20], [21], [22] have been developed for community detection. However, the performance of community detection methods based on modularity Q is often constrained by the resolution limitation [23]. The resolution limitation arises because modularity Q is defined based on heuristic assumptions, which may not align with the actual community structure.

Several methods [24], [25], [26] have been proposed to address the resolution limitation by introducing a resolution parameter into the modularity definition. Another common approach to address the resolution limitation is local community detection, which focuses on identifying the local community structure around a given node. This approach is less susceptible to interference from other communities or the overall network size. However, local community detection methods based on local modularity also face limitations due to the intrinsic limitations of local modularity, leading to precision limitation. Therefore, local community detection only partially mitigates the resolution limitation.

B. Local Community Detection

As network scales continue to grow, the computational time and memory requirements of many classical global community detection methods increase substantially [26]. In some cases, the focus shifts to a specific community. Local community detection methods can identify the community to which a given node belongs. Moreover, local community detection is a widely used strategy to address the resolution limitation. As a result, local community detection has become a key focus in current research.

The concept of local community detection was first proposed by Clauset [14]. He subsequently defined local modularity R and proposed a method for detecting local communities based on R, hereafter referred to as the R-based method. This method selects the node that maximizes the gain in local modularity R to join the local community at each iteration, until local modularity R no longer yields a gain. Luo et al. [27] proposed another typical local modularity M and a local community detection method based on modularity M optimization, hereafter referred to as the M-based method. Both the R-based and M-based methods are sensitive to the initial node's position, often leading to reduced precision and the generation of outliers [18]. Over the past few years, numerous other local community detection methods have been proposed [28], [29], [30], [31], [32], [33], [34]. These methods improve performance by refining the modularity function or designing expansion rules. For example, Luo et al. [28] proposed the LCDNN method, which utilizes the concept of NGC nodes as described in [35]. The LCDNN method requires that: (1) The fuzzy relation value of a node to its NGC node is the largest among all candidate nodes. (2) If the node is added to the local community, the mean value of the fuzzy relationships among the nodes in the community will not decrease.

C. Precision Limitation in Local Community Detection

Typically, local community detection methods based on optimization iteratively add nodes to the community in a greedy manner until the local optimization function reaches its maximum. Therefore, whether a node joins the local community is determined by the characteristics of the local optimization function. In such cases, precision limitation appears.

To address precision limitation, some methods based on local modularity adopt a strategy similar to community detection methods by introducing a hyperparameter λ to detect local communities with different scales. For example, the DMF_R method and the DMF_M method [29] partition the local community expansion process into three stages: the initial stage, the intermediate stage and the final stage. Each stage employs different objective functions to determine whether nodes should join the local community.

Based on the above observations, in this paper, we propose a new local community detection method called FG-based. The method mitigates the impact of noise on community detection by leveraging the relationship characteristics of nodes to construct folded subgraphs. Furthermore, the FG-based method employs a three-stage expansion strategy, enabling nodes of different types to join local communities at distinct stages. This method aims to address precision limitation.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.

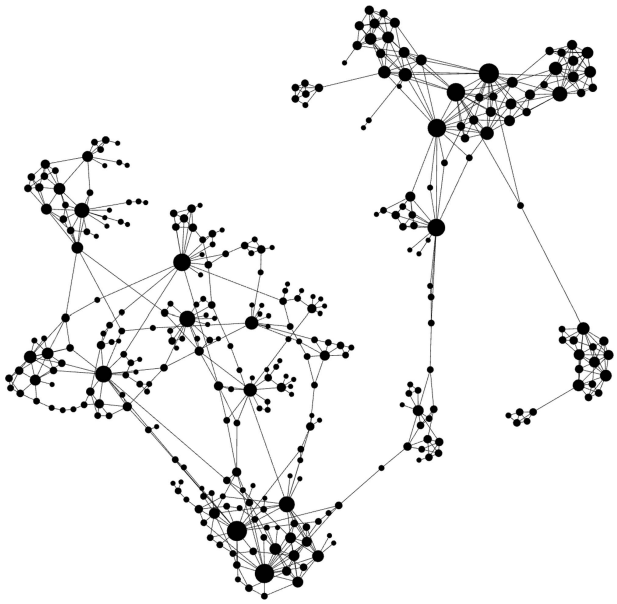


Fig. 1. An example showing following relationships among nodes in a network.

III. The Folded Subgraph Method

A. Network Analysis

Many studies have shown that the degree distribution of nodes in networks follows a power-law distribution. In a real world network, most nodes have a relatively small degree, but a few nodes will have very large degree, being connected to many other nodes. For example, Fig. 1 illustrates the connections within a community in the Amazon dataset.

From the figure, it is evident that each node belongs to a tightly connected small group within the local community. These small groups are then interconnected to form the local community. However, due to such a structure, the local modularity is difficult to generate gains in the process of local community expansion, resulting in unsatisfactory performance of traditional local community detection methods based on local modularity. Additionally, the lack of information about neighboring nodes and the presence of noise in the network topology often lead to suboptimal results.

In community detection for large-scale networks, following relationships among nodes are common. For example, a node with a degree of 1 naturally forms a following relationship with its neighbor node. Similarly, for nodes with low degrees, we can use their topological information to identify similar nodes and form a node set. The subgraph formed by the node set is expected to have a relatively high link density. A high link density in the subgraph indicates that the nodes are tightly connected, making it more likely that they belong to the same community and maintain a following relationship.

Therefore, we can use link density as a basis to identify clique structures and compress each clique into a single node for subsequent expansion. On the one hand, the link density inside the clique is relatively large, which ensures that the nodes in the same clique belong to the same community. On the other hand, this approach enhances the extraction of useful information while reducing noise. Based on this, we propose the Folded Subgraph method.

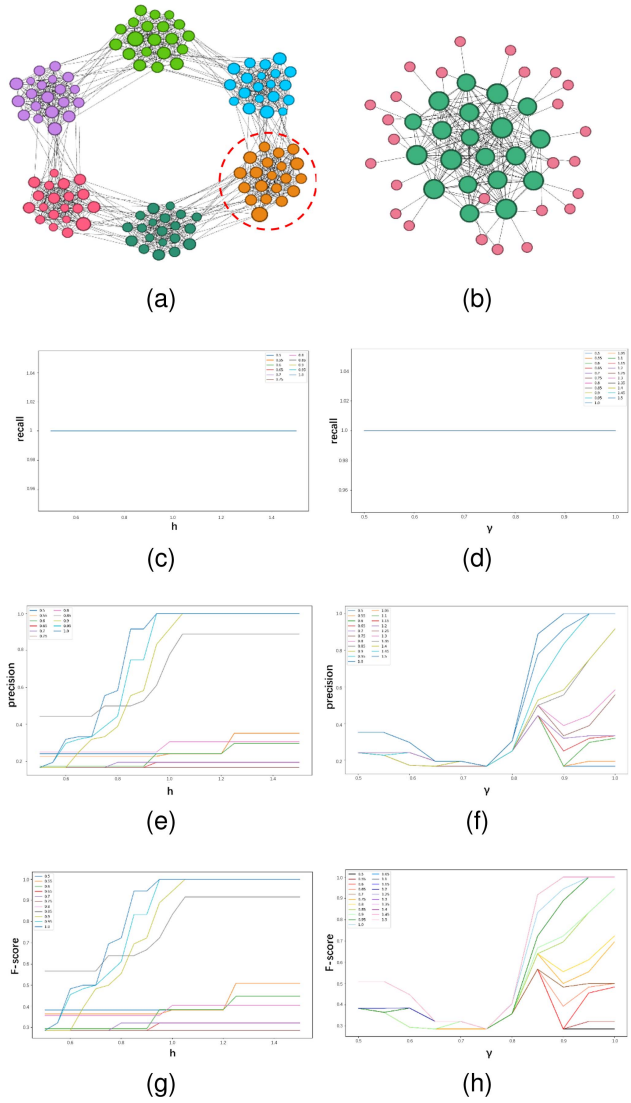


Fig. 2. Experimental results on hyperparameters of the community-loop network. (a) The basic structure of the network. (b) The basic structure of an community(red circle marking) in the network, where the green is the local community and the red is the set of external nodes directly connected to the local community. (c) and (d) show the trend of recall with h and γ, respectively. (e) and (f) show the trend of precision with h and γ, respectively. (g) and (h) show the trend of F-score with h and γ, respectively.

B. Method Introduction

First we define some important set notation:

1) The set $\mathbb{F}$ denotes the set of nodes corresponding to the folded subgraph.
2) The set $\mathbb{R}$ denotes the set of alternative nodes. The set $\mathbb{R}$ is the neighbor nodes set of the set $\mathbb{F}$, that is:

$$\mathbb{R} = \bigcup_{v \in \mathbb{F}} N(v) - \mathbb{F}$$

where $N(v)$ denotes the neighbor nodes set of node v.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.

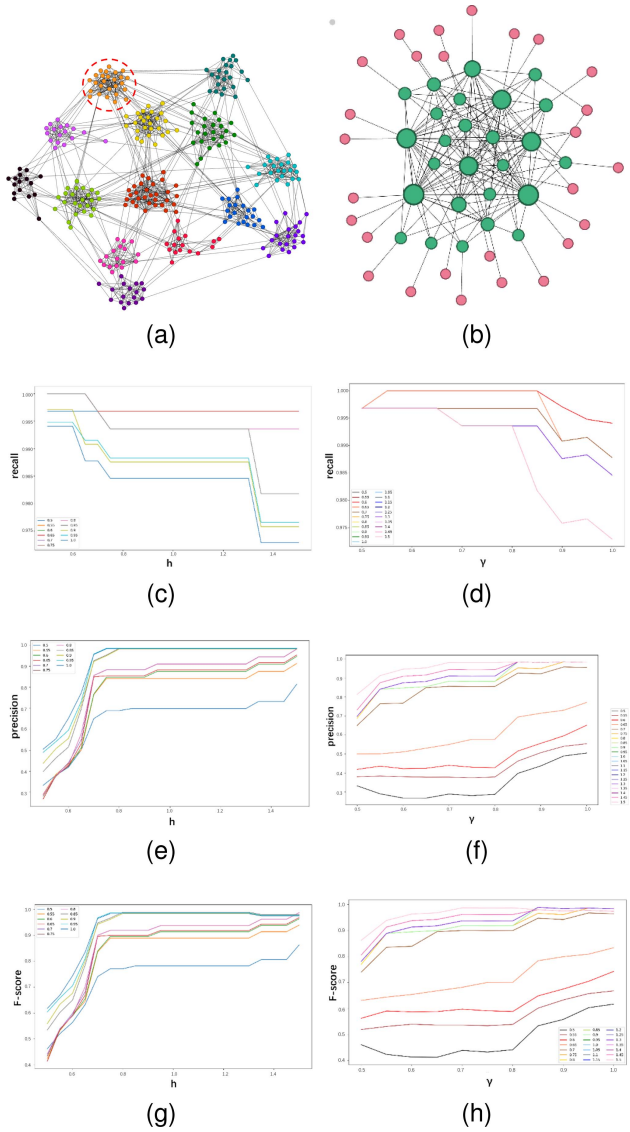


Fig. 3. Experimental results on hyperparameters of the LFR(0.46) network. (a) The basic structure of the network. (b) The basic structure of an community(red circle marking) in the network. (c) and (d) show the trend of recall with h and γ, respectively. (e) and (f) show the trend of precision with h and γ, respectively. (g) and (h) show the trend of F-score with h and γ, respectively.

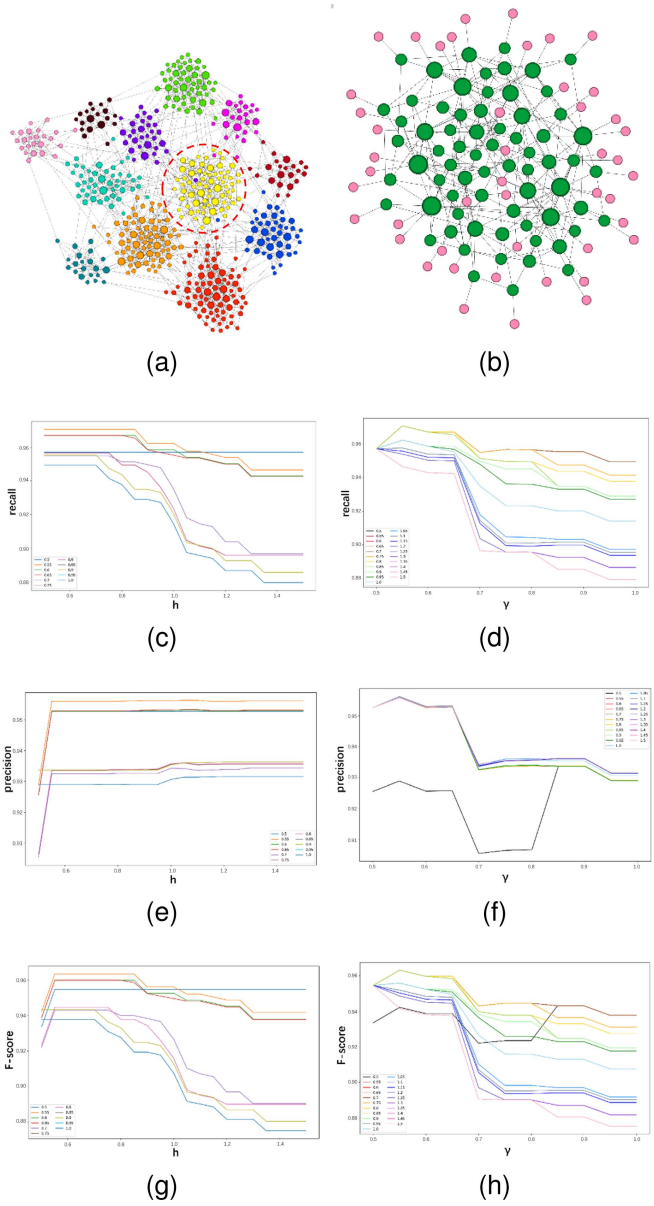


Fig. 4. Experimental results on hyperparameters of the LFR(0.18) network. (a) The basic structure of the network. (b) The basic structure of an community(red circle marking) in the network. (c) and (d) show the trend of recall with h and γ, respectively. (e) and (f) show the trend of precision with h and γ, respectively. (g) and (h) show the trend of F-score with h and γ, respectively.

To form the folded subgraph, we employ a greedy strategy. At each step, we select the node with the most edges connected to $\mathbb{F}$ and add it to $\mathbb{F}$. If multiple nodes have the maximum number of edges, the node with the lowest degree will be selected. Based on the previous analysis, we aim for the link density of the folded subgraph to be sufficiently high. To achieve this, we introduce the parameter γ. We require that the link density of the folded subgraph must be greater than or equal to γ. Thus, we continue adding nodes to the set $\mathbb{F}$, using the greedy strategy until the link density drops below γ. By definition, γ should be a value between 0 and 1. The value of γ dictates the link strength of the folded subgraph. Experimental analysis suggests that γ should be assigned a relatively high value, preferably above 0.5.

Here, we introduce the proposed method to get folded subgraph, which is shown in Algorithm 1.

IV. Proposed Method

In this section, we focus on the extension mechanism of the FG-based method. The community expansion in the FG-based method consists of three stages: the folding stage, the modularity-based expansion stage and the relaxation stage. These stages operate sequentially in the local community detection process, enabling the FG-based method to identify and incorporate relevant nodes at each stage, thereby improving recall and precision. For clarity, we first define the following mathematical notations:

1) $\mathbb{C}$ denotes the local community obtained by the current extension.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.

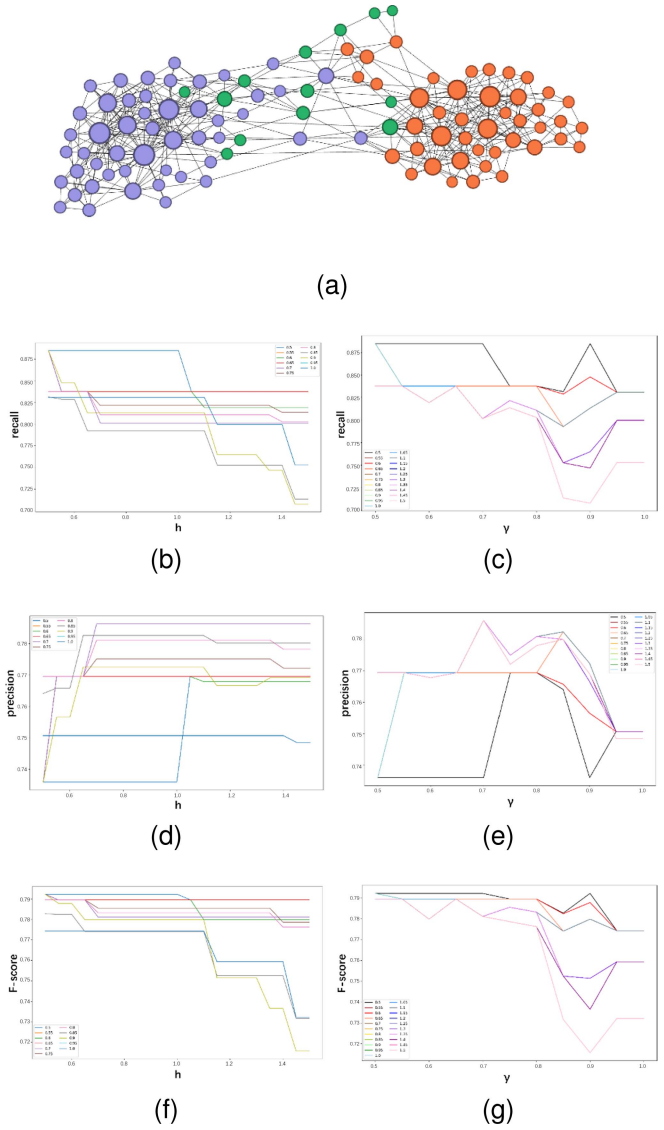


Fig. 5. Experimental results on hyperparameters of the Polbooks dataset. (a) The basic structure of the network. (b) and (c) show the trend of recall with h and γ, respectively. (d) and (e) show the trend of precision with h and γ, respectively. (f) and (g) show the trend of F-score with h and γ, respectively.

2) $\mathbb{A}$ denotes an alternative set of nodes to be added to the community. The set $\mathbb{A}$ is the neighbor nodes set of the set $\mathbb{C}$, that is:

$$\mathbb{A} = \bigcup_{v \in \mathbb{C}} N(v) - \mathbb{C}$$

3) G is the current local network, which is composed of nodes in the set $\mathbb{C}$ and the set $\mathbb{A}$.

A. The Folding Stage

For each node v in the set $\mathbb{A}$, Algorithm 1 is used to construct its folded subgraph $\mathbb{F}(v)$. The folded subgraph exhibits high link density. Thus, if more than half of the nodes in the folded subgraph belong to the local community $\mathbb{C}$, all nodes in the folded subgraph are directly included in $\mathbb{C}$, improving the efficiency of local community expansion. Algorithm 2 describes the core process of local expansion during the folding stage.

Algorithm 1: The Folded Subgraph Method.

Require: Local network: G; The starting node: v; Parameters: γ
Ensure: Set $\mathbb{F}$
1: Initial $\mathbb{F} = \{\mathrm{v}\}$, $\mathbb{R} = N(v)$
2: **while** $\mathbb{R} \neq \emptyset$ **do**
3: **for** each node v in $\mathbb{R}$ **do**
4: Count the number of edges between node v and the set $\mathbb{F}$
5: **end for**
6: Select the node with the most connected edges. If there are more than one node, select the node with the smallest degree, denoted as w.
7: **if** the link density of the set $\mathbb{F}$ after adding node $\mathrm{w} < \gamma$ **then**
8: break
9: **end if**
10: Add node w to the set $\mathbb{F}$
11: Update the set $\mathbb{R}$
12: **end while**

Algorithm 2: The Folding Stage.

Require: Local network: G; Inital local community: $\mathbb{C}$; The alternative set $\mathbb{A}$
Ensure: Local community: $\mathbb{C}$
1: **while** True **do**
2: **for** each node v in $\mathbb{A}$ **do**
3: Use Algorithm 1 to obtain the folded subgraph $\mathbb{F}(v)$
4: **if** $\mid \mathbb{F}(v) \cap \mathbb{C} \mid \geq 0.5* \mid \mathbb{F}(v) \mid$ **then**
5: Add $\mathbb{F}(v)$ to the set $\mathbb{C}$
6: Update the alternative set $\mathbb{A}$
7: **end if**
8: **end for**
9: **if** no new nodes can be added to $\mathbb{C}$ **then**
10: break
11: **end if**
12: **end while**

B. The Modularity-Based Expansion Stage

The modularity-based expansion stage is similar to the traditional local community detection methods based on local modularity optimization. We use the local modularity M as the optimization function.

$$M = \frac{e_{in}}{e_{out}} \tag{1}$$

where e_{in} denotes the number of edges within the community and e_{out} denotes the number of edges between the community and the outside.

We adopt a greedy strategy. Algorithm 1 is used to construct the corresponding folded subgraph $\mathbb{F}(v)$, and the local modularity M is computed after adding $\mathbb{F}(v)$ to the local community $\mathbb{C}$.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.

Algorithm 3: The Modularity-Based Expansion Stage.

Require: Local network: G; Inital local community: $\mathbb{C}$; The alternative set $\mathbb{A}$
Ensure: Local community: $\mathbb{C}$
while $\mathbb{A} \neq \emptyset$ **do**
  for each node v in $\mathbb{A}$ **do**
    Use Algorithm 1 to get the folded subgraph $\mathbb{F}(v)$
    Calculate ΔM according to formula 5
  end for
  $\Delta M_{best} = \max_{v \in \mathbb{A}} \Delta M_{\mathbb{F}(v)}$ and $\mathbb{F}(v_{best})$ is its corresponding folded subgraph
  if $\Delta M_{best} < 0$ **then**
    break
  end if
  Add $\mathbb{F}(v_{best})$ to the set $\mathbb{C}$
  Update e_{in}, e_{out} and $\mathbb{A}$
end while

The folded subgraph yielding the largest gain in local modularity M is selected and added to $\mathbb{C}$ until the local modularity gain ΔM becomes negative.

For the folded subgraph $\mathbb{F}(v)$, we can directly update e_{in} and e_{out} during local community expansion, so as to speed up the calculation of the local modularity M gain. The specific calculation process is as follows:

1) x: The number of new edges between $\mathbb{F}(v)$ and the local community $\mathbb{C}$

$$x = \sum_{i \in \mathbb{F}(v) - \mathbb{C}} \mid N(i) \cap \mathbb{C} \mid \tag{2}$$

2) y: The number of edges in e_{out} associated with $\mathbb{F}(v)$

$$y = \sum_{i \in \mathbb{F}(v) \cap \mathbb{A}} \mid N(i) \cap \mathbb{C} \mid \tag{3}$$

3) z: The number of edges between $\mathbb{F}(v)$ and the outside of the local community $\mathbb{C}$

$$z = \mid \sum_{i \in \mathbb{F}(v) - \mathbb{C}} degree(i) \mid - x - 0.5 * \sum_{i \in \mathbb{F}(v) - \mathbb{C}} \mid N(i) \cap (\mathbb{F}(v) - \mathbb{C}) \mid \tag{4}$$

Thus, if the folded subgraph $\mathbb{F}(v)$ joins the local community $\mathbb{C}$, then the local modularity M gain obtained is as follows.

$$\Delta M_{\mathbb{F}(v)} = \frac{e_{in} + x}{e_{out} + z - y} - \frac{e_{in}}{e_{out}} \tag{5}$$

Algorithm 3 is our basic process for local expansion in the modularity-based expansion stage.

C. The Relaxation Stage

In the first two stages, newly added nodes must be tightly connected to the local community, with sparse external links. Unlike the strong fusion effect in the first two stages, the relaxation stage involves selecting folded subgraphs with weaker connections to the local community and adding them to the local community. This allows the local community to extend further beyond the first two stages through relaxation. We represent the relationship between folded subgraphs and local communities from two aspects, denoted as $f_1(v)$ and $f_2(v)$ respectively. Their definitions are provided in formulas (6) and (7).

$$f_1(v) = \frac{\mid \bigcup_{i \in \mathbb{F}(v)} N(i) \cap \mathbb{C} \mid}{\mid \mathbb{C} \mid} \tag{6}$$

where $N(i)$ denotes the neighbor nodes set of node i and $\mid \bigcup_{i \in \mathbb{F}(v)} N(i) \cap \mathbb{C} \mid$ denotes the intersection of the neighbor nodes set of $\mathbb{F}(v)$ and the local community $\mathbb{C}$. Analogously to $f_1(v)$, we give the definition of $f_2(v)$.

$$f_2(v) = \frac{\mid \bigcup_{i \in \mathbb{F}(v)} NN(i) \cap \mathbb{C} \mid}{\mid \mathbb{C} \mid} \tag{7}$$

where $NN(i)$ denotes the second-order neighbor nodes set of node i, that is, $NN(i) = \{v_k | v_k \in N(N(i))\}$. $\mid \bigcup_{i \in \mathbb{F}(v)} NN(i) \cap \mathbb{C} \mid$ denotes the intersection of the second-order neighbor nodes set of $\mathbb{F}(v)$ and the local community $\mathbb{C}$.

$f_1(v)$ measures the closeness between the folded subgraph and the local community using link density. A higher number of connections between the folded subgraph and the local community indicates greater closeness. $f_2(v)$ is similar to $f_1(v)$ but uses second-order neighbors to represent closeness. After adding the folded subgraph, the modularity-based expansion stage utilizes its neighbor nodes. $f_2(v)$ ensures strong connections between the neighbor nodes of the folded subgraph and the local community, facilitating smoother expansion in the modularity-based stage.

The relaxation stage employs a weak fusion strategy, adding the folded subgraph with the maximum value of $f(v) = f_1(v) + f_2(v) * \beta$ to the local community based on multi-dimensional similarity. Here, β referred to as the relaxation factor, represents the contribution ratio of $f_1(v)$ and $f_2(v)$. Experimental results show that $\beta = 0.5$ yields optimal performance, and thus we set $\beta = 0.5$ in this paper.

In the later stages of local community expansion, when the community has grown to a certain size, unrestricted values of f may allow folded subgraphs with low f values to be added to the community. Folded subgraphs with low f values are weakly correlated with the local community, potentially leading to incorrect expansions. Therefore, we introduce the hyperparameter h.On the basis of identifying the folding subgraph with maximum value of f, only if $f \geq h$, can we choose to add it to the local community. The hyperparameter h ensures that folded subgraphs added during the relaxation stage exhibit both high membership and significant influence. Since $f_1(v)$ and $f_2(v)$ range from 0 to 1, h ranges from 0 to 1.5.

Algorithm 4 is our basic process for local expansion in the relaxation stage.

Finally, we give the overall process of FG-based method, namely Algorithm 5.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.

Algorithm 4: The Relaxation Stage.

Require: Local network: G; Inital local community: $\mathbb{C}$; The alternative set $\mathbb{A}$; hyperparameter: h
Ensure: Local community: $\mathbb{C}$
1: **for** each node v in $\mathbb{A}$ **do**
2: Use Algorithm 1 get the folded subgraph $\mathbb{F}(v)$
3: Calculate $f_1(v)$ and $f_2(v)$ according to formula 6 and 7
4: **end for**
5: $f_{best} = \max_{v \in \mathbb{A}} f(v)$, and $\mathbb{F}(v_{best})$ is its corresponding folded subgraph
6: **if** $f_{best} \geq h$ **then**
7: Add $\mathbb{F}(v_{best})$ to the set $\mathbb{C}$
8: Update e_{in}, e_{out} and $\mathbb{A}$
9: **end if**

Algorithm 5: FG-Based Method.

Require: Local network:G;Inital local community: $\mathbb{C}$; The alternative set $\mathbb{A}$
Ensure: Local community: $\mathbb{C}$
1: Initialize nums1 = nums2 = the number of nodes in inital local community $\mathbb{C}$
2: **while** $\mathbb{A} \neq \emptyset$ **do**
3: Utilize Algorithm 2 for the local extension of the folding stage
4: Utilize Algorithm 3 for the local extension of the modularity-based expansion stage
5: Let nums1 = the number of nodes in local community $\mathbb{C}$
6: **if** $nums1 = nums2$ **then**
7: break
8: **end if**
9: Utilize Algorithm 4 for the local extension of the relaxation stage
10: Let nums2 = the number of nodes in local community $\mathbb{C}$
11: **end while**

V. Experiment

A. Dataset

We test our method on four real datasets and nine synthetic LFR networks. Four real datasets are Karate [36], Dolphins [37], US politics books [38] and Amazon [39]. The LFR artificial network was designed by Lancichinetti et al. in [40]. LFR artificial network has many network structure parameters. N is the number of nodes in the network, k is the average degree of the nodes in the network, $maxk$ is the maximum degree of the nodes in the network, μ is the mixing coefficient, t_1 is the minus exponent for the degree sequence, t_2 is the minus exponent for the community size distribution, $minc$ is the lower limit of the community size, $maxc$ is the upper limit of the community size, and on and om are coefficients on overlapping nodes. We set a total of nine artificial datasets, of which corresponding parameters are shown in Table I.

B. Comparisons

To evaluate the performance of our proposed method, we compare it with several classical local community detection methods based on local modularity optimization. Table II summarizes the details of the compared methods.

To evaluate our proposed method, we compare it with several classical local community detection methods. Table II details the methods used for comparison.

C. Evaluation Metrics

We use recall, precision and F-score as evaluation metrics to assess the performance of local community detection methods. For each node in the network, we identify its local community and compute recall, precision, and F-score. The final performance of the method is obtained by averaging recall, precision, and F-score across all nodes.

$$recall = \frac{\mid C_{found} \cap C_{true} \mid}{\mid C_{true} \mid}$$

$$precision = \frac{\mid C_{found} \cap C_{true} \mid}{\mid C_{found} \mid}$$

$$F - score = \frac{2 * precision * recall}{precision + recall}$$

Where C_{found} denotes the local community obtained by the method, and C_{true} denotes the true local community.

D. Results on Real Datasets

The results on real datasets show that the FG-based method significantly improves recall compared to other methods. Specifically, as the FG-based method uses local modularity M during the modularity-based expansion stage, we focus on comparing it with the M-based method. Recall on the four datasets is increased by 55.23%, 175.37%, 82.92%, 26.6% respectively. Precision is also improved on the karate dataset, and in the rest of the datasets only drops of 0.01 to 0.03. With a significant improvement in recall and only a minor loss in precision, the F-score is greatly enhanced across all datasets. Compared to all methods, the FG-based method achieves the highest F-score.

E. Results on LFR Datasets

For the LFR1-5 dataset, μ ranges from 0.1 to 0.5. As μ increases, the community structure becomes less distinct, reducing the effectiveness of the methods. Compared to the R-based, M-based and LQ-based methods, the FG-based method achieves the best recall and precision. Since the FG-based method also uses local modularity M for expansion, the folded subgraph and the three-stage expansion mechanism enhance the performance of local modularity M in identifying communities across varying mixing coefficients μ. Compared to the LCDNN and FSR methods, precision slightly decreases when the community structure is clear, but recall is significantly improved. However, when μ exceeds 0.3, the performance of the FG-based method declines rapidly. This is because, with high μ values, the community

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PARAMETER SETTINGS IN LFR NETWORKS

	LFR1	LFR2	LFR3	LFR4	LFR5	LFR6	LFR7	LFR8	LFR9
N	200	200	200	200	200	1000	1000	500	1000
k	7	7	7	7	7	15	15	7	7
maxk	15	15	15	15	15	50	50	15	15
μ	0.1	0.2	0.3	0.4	0.5	0.1	0.1	0.1	0.1
t1	2	2	2	2	2	2	2	2	2
t2	1	1	1	1	1	1	1	1	1
minc	10	10	10	10	10	10	10	10	10
maxc	80	80	80	80	80	50	300	80	80
on	0	0	0	0	0	0	0	0	0
om	0	0	0	0	0	0	0	0	0

TABLE II
INFORMATION OF FIVE COMPARISON METHODS

Method	Underlying idea	Year	Ref.
M-based	use local modularity M	2006	[14]
R-based	use local modularity R	2017	[27]
LQ-based	use local modularity LQ	2018	[15]
LCDNN	use NGC nodes	2020	[28]
FSR	use fuzzy similarity relationship	2020	[11]

structure becomes unclear, leading to unstable gains in local modularity M and the inclusion of incorrect nodes. Therefore, when the community structure is not obvious, the FG-based method will be limited due to the problem of the local modularity M itself.

The LFR6 and LFR7 datasets mainly differ in community size. From the results, the recognition is basically completely correct. Therefore, FG-based method can be applied to local community detection of different sizes of communities. LFR1, LFR8 and LFR9 differ primarily in the number of nodes in the network. Our proposed method is robust to network scale and outperforms other methods.

F. Operating Efficiency and Initial Node Sensitivity

In Table V, Time(s) represents the average time taken to generate local communities with any initial node from LFR1-LFR9 datasets. Recall, Precision, and F-score represent the average values for local communities across all nodes in the LFR1-LFR9 datasets, and Var indicates the variance of F-score values across these nodes. Although the FG-based method takes more time on average than the M-based and R-based methods, it significantly improves Recall, Precision, and F-score. Among all methods, the FG-based method has the lowest Var value, indicating lower sensitivity to initial values.

VI. DISCUSSION

A. The Necessity and Effectiveness of the Three-Stage Extension

Compared to traditional local community detection methods relying on local modularity optimization, we propose a three-stage expansion strategy. This strategy combines the folded subgraph strategy (comprising the folded subgraph method and the folding stage expansion) with the relaxation stage expansion. To demonstrate the necessity and effectiveness of the relaxation and folded subgraph strategies in the FG-based method, we conducted experiments on LFR datasets.

We now introduce the pure expansion method. The pure expansion method omits the folded subgraph approach and retains only the expansion mechanism. Since it excludes the folded subgraph strategy, it essentially combines the M-based method with the relaxation stage.

In the experiment, we compared the performance of the M-based method with the pure extension method to demonstrate the necessity of the relaxation stage. We also compared the pure extension method with the FG-based method to illustrate the necessity of the folded subgraph strategy.

When the community structure is less distinct (when μ is greater than 0.3), the M-based, pure expansion, and FG-based methods perform similarly. The pure expansion method significantly boosts recall compared to the M-based method but at the cost of precision. As a result, the overall improvement in recall is limited. This suggests that while the relaxation method can include many nodes, it negatively impacts modularity-based expansion and reduces precision. In contrast, the FG-based method leverages folded subgraphs to provide more local information, thereby improving precision.

When the community structure is clear (when μ is less than or equal to 0.3), the pure expansion method improves both recall and precision compared to the M-based method, particularly in recall, demonstrating the effectiveness of the relaxation strategy in significantly mitigating the precision limitation. The relaxation strategy not only adds nodes closely related to the local community but also enhances recall by leveraging local modularity M. Thus, the relaxation strategy is both effective and essential.

When the community structure is clear (when μ is less than or equal to 0.3), the FG-based method further improves recall and precision compared to the pure expansion method. The higher recall indicates that the folded subgraph incorporates more fringe nodes into the local community. The improved precision suggests that the folded subgraph offers additional local information, reducing extension errors. Thus, the folded subgraph strategy is also effective and essential.

B. Selection of Parameters

Our proposed method includes two key hyperparameters: the parameter γ in the folded subgraph method and the parameter h in the community expansion stage. The choice of

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARISONS ON TRUE DATASETS

Datasets		R-based	M-based	LQ-based	LCDNN	FSR	FG-based
Karate	Recall	0.5527	0.6442	0.7032	**1.0**	0.6091	**1.0**
	Precision	0.9088	0.8905	0.8905	0.9477	**1.0**	**1.0**
	F-score	0.6474	0.7007	0.7598	0.9723	0.6922	**1.0**
Dolphins	Recall	0.3230	0.3464	0.5546	0.6594	0.4919	**0.9539**
	Precision	0.9648	0.9667	0.9528	0.9693	**1.0**	0.9413
	F-score	0.4505	0.4685	0.6489	0.7558	0.5868	**0.9466**
US politics books	Recall	0.4410	0.4837	0.5938	0.7679	0.8316	**0.8848**
	Precision	**0.7752**	0.7664	**0.7729**	0.7577	0.7379	0.7359
	F-score	0.5030	0.5257	0.6178	0.7463	0.7543	**0.7923**
Amazon	Recall	0.5688	0.5993	0.6382	0.7424	0.6894	**0.7587**
	Precision	**0.9281**	0.9137	0.9273	0.9133	0.9045	0.9004
	F-score	0.6322	0.6465	0.7013	0.7546	0.7601	**0.7783**

TABLE IV
COMPARISONS ON LFR DATASETS

Datasets		R-based	M-based	LQ-based	LCDNN	FSR	FG-based
LFR1	Recall	0.6794	0.6834	0.6952	0.8308	0.6184	**0.9713**
	Precision	0.8617	0.8619	0.8536	0.9424	**1.0**	0.9319
	F-score	0.7087	0.7119	0.7302	0.8566	0.6698	**0.9500**
LFR2	Recall	0.4367	0.4436	0.6137	0.6945	0.5472	**0.9097**
	Precision	0.6843	0.6839	0.6985	**0.8775**	0.816	0.8635
	F-score	0.4867	0.4908	0.6249	0.7394	0.5317	**0.8775**
LFR3	Recall	0.3	0.3008	0.4993	0.3936	0.3311	**0.6708**
	Precision	0.4947	0.4939	0.498	0.661	**0.8441**	0.5637
	F-score	0.3297	0.3294	0.4614	0.4571	0.3362	**0.5781**
LFR4	Recall	0.2398	0.2414	0.3478	0.4685	0.363	**0.794**
	Precision	0.3453	0.3450	0.3163	0.4947	**0.5108**	0.2559
	F-score	0.2214	0.2208	0.2696	**0.4059**	0.1831	0.3326
LFR5	Recall	0.2266	0.2287	0.3414	0.304	0.1113	**0.8852**
	Precision	0.3148	0.3138	0.2986	0.3527	**0.7676**	0.2045
	F-score	0.2171	0.2159	0.2819	0.2563	0.1488	**0.3009**
LFR6	Recall	0.8109	0.8109	0.8147	**1.0**	**1.0**	**1.0**
	Precision	0.8256	0.8256	0.8271	**1.0**	**1.0**	**1.0**
	F-score	0.8113	0.8113	0.8147	**1.0**	**1.0**	**1.0**
LFR7	Recall	0.6421	0.6421	0.651	0.9528	0.9419	**0.9937**
	Precision	0.8274	0.8274	0.8278	0.999	**1.0**	0.9923
	F-score	0.6547	0.6547	0.6646	0.9598	0.9548	**0.9929**
LFR8	Recall	0.5408	0.5421	0.5694	0.5619	0.4887	**0.9707**
	Precision	0.8235	0.8236	0.8236	0.9469	**1.0**	0.956
	F-score	0.5831	0.584	0.6196	0.6411	0.9548	**0.9632**
LFR9	Recall	0.6221	0.6247	0.6328	0.5631	0.5504	**0.9588**
	Precision	0.824	0.8244	0.8188	0.9576	**1.0**	0.9302
	F-score	0.652	0.6537	0.6664	0.7847	0.613	**0.9396**

TABLE V
OPERATING EFFICIENCY AND INITIAL NODE SENSITIVITY

	R-based	M-based	LQ-based	LCDNN	FSR	FG-based
Time(s)	0.0291	0.0038	0.0515	0.1331	0.0702	0.0513
Recall	0.6049	0.6063	0.6406	0.7795	0.6957	0.9520
Precision	0.7620	0.7620	0.7597	0.9102	0.9528	0.8810
F-score	0.6227	0.6234	0.6509	0.8018	0.7163	0.8938
Var	0.1458	0.1185	0.1398	0.0514	0.0764	0.0226

hyperparameters significantly affects the performance of the FG-based method. Thus, we analyze hyperparameter selection using diverse datasets, offering guidance for future applications of the method. The datasets differ primarily in the average link density within communities and the overall network density. The number of nodes in datasets is on the order of hundreds. Networks with hundreds of nodes effectively demonstrate the impact of hyperparameter selection on method performance while remaining suitable for visualization.

1) Community-Loop Network [41]: In each community-loop network, r communities are connected sequentially, with each community containing n_c nodes. Nodes are connected only to nodes within the same community or two adjacent communities. We conducted an experiment on a community-loop network with an average intra-community link density of 0.8096 and a network density of 0.1461. Due to the high intra-community link density, all nodes within the community are easily identified, resulting in a recall of 1. In terms of precision, with the fixed parameter γ, precision keeps increasing as the parameter h increases. This is because higher values of h restrict node inclusion during the relaxation stage, reducing the likelihood of incorrect node additions. When h is fixed, precision shows an overall upward

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
EXPERIMENT FOR VERIFING THREE-STAGE EXTENSION

Datasets		M-based	the pure extension method	FG-based
LFR1	Recall	0.6834	0.8644	**0.9713**
	Precision	0.8619	0.8565	**0.9319**
	F-score	0.7119	0.8561	**0.9500**
LFR2	Recall	0.4436	0.7869	**0.9097**
	Precision	0.6839	0.7406	**0.8635**
	F-score	0.4908	0.7487	**0.8775**
LFR3	Recall	0.3008	0.6001	**0.6708**
	Precision	0.4939	0.4642	**0.5637**
	F-score	0.3294	0.4949	**0.5781**
LFR4	Recall	0.2414	**0.9147**	0.794
	Precision	**0.3450**	0.2034	0.2559
	F-score	0.2208	0.3097	**0.3326**
LFR5	Recall	0.2287	0.8348	**0.8852**
	Precision	**0.3138**	0.2119	0.2045
	F-score	0.2159	0.2932	**0.3009**
LFR6	Recall	0.8109	0.8308	**1.0**
	Precision	0.8256	0.83	**1.0**
	F-score	0.8113	0.8296	**1.0**
LFR7	Recall	0.6421	0.8108	**0.9937**
	Precision	0.8274	0.8236	**0.9923**
	F-score	0.6547	0.8111	**0.9929**
LFR8	Recall	0.5421	0.8637	**0.9707**
	Precision	0.8236	0.8448	**0.956**
	F-score	0.584	0.8522	**0.9632**
LFR9	Recall	0.6247	0.836	**0.9588**
	Precision	0.8244	0.8356	**0.9302**
	F-score	0.6537	0.8292	**0.9396**

trend as γ increases, with some fluctuations in the middle. Fluctuations mainly occur when h is small. In such cases, a large γ limits the information provided by the folded subgraph (which may shrink to a single node), making the method more susceptible to network noise and reducing precision.

2) LFR(0.46) Dataset: This LFR dataset contains 300 nodes, with a network density of 0.0352 and an average intra-community link density of 0.4618. The experimental results show clear upward or downward trends in each metric as the parameters change. Fixed the parameter γ, recall decreases and precision increases as h increases. Fixed the parameter h, recall decreases and precision increases as γ increases. Due to the high intra-community link density, recall changes minimally, while precision changes significantly, causing the F-score and precision curves to follow similar trends.

3) LFR(0.18) Dataset: This LFR dataset contains 500 nodes, with a network density of 0.013 and an average intra-community link density of 0.1872. When γ is fixed, recall decreases as h increases, while precision improves. When h is fixed, recall and precision generally decrease as γ increases. This occurs because a large γ results in a smaller folded subgraph, providing limited information. Additionally, for nodes on community boundaries, the link density within the community is similar to that outside the community.As a result, the FG-based method becomes more susceptible to noise, reducing precision.

4) Polbooks Dataset: The Polbooks dataset has a network density of 0.0808 and an average intra-community link density of 0.1558.

When γ is fixed, recall decreases as h increases, while precision improves. When h is taken as a relatively high value, precision may have a small decline. This occurs because, at high h values, some correct nodes fail to join the local community during the relaxation stage. When h is fixed, recall generally decreases as γ increases, while precision initially increases and then decreases with higher γ values. Precision initially improves because higher γ values increase the likelihood that nodes in the folded subgraph belong to the same community. The subsequent decline of precision is due to the reduced information provided by the folded subgraph and the effect of noise.

The parameter γ determines the strictness for nodes to join a folded subgraph. Higher γ values require greater link density in the folded subgraph, making it harder for nodes to join. The parameter h determines the strictness for nodes to join the local community during the relaxation stage. Higher h values impose stricter relevance requirements between nodes and the local community, making it harder for nodes to join. The performance of the FG-based method depends heavily on the choice of parameters γ and h. Based on the experimental results, appropriate values for h and γ can be selected according to the average intra-community link density for different datasets. This can be categorized into two main scenarios.

As for parameter h, the purpose of the relaxation phase is to join nodes in the same community that have not yet been extended. The high density of links within the community ensures that the f value calculated by such nodes should also be a large value, so choosing a higher h can improve the value of precision without losing the value of recall. To sum up, it is recommended to use larger γ and larger h.

For communities with relatively high intra-community link density, the trends of F-score and Precision are consistent, while Recall remains largely stable as parameters h and γ vary. This is because the community is tightly connected and easier to identify. In this case, small parameter values make it difficult to clearly define community boundaries. For instance, the high

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.

intra-community link density means that even if external nodes join the folded subgraph, the impact on link density is minimal. Thus, a higher γ is needed to minimize the inclusion of incorrect nodes. For the parameter h, the relaxation stage aims to include nodes within the same community that were not previously added. The high intra-community link density ensures that such nodes have high f values, so a higher h can improve precision without sacrificing recall. In summary, larger values of γ and h are recommended.

For communities with relatively low intra-community link density, the trends of F-score and Recall are consistent. In such cases, expansion based on local modularity gain becomes challenging. Thus, a smaller h is typically chosen to ensure high recall. Similarly, γ should not be too large, typically ranging from 0.55 to 0.8.

VII. Conclusion

Local community detection methods rely on local information, often affected by network noise and redundant data, resulting in lower-quality local communities. Additionally, local community detection methods based on local modularity optimization suffer from the precision limitation. In this paper, we propose the FG-based method to address these challenges. The FG-based method leverages network characteristics, enabling nodes with smaller degrees to join the local community through their influential neighbors. Thus, for a given node, we identify similar nodes to construct a folded subgraph. A three-stage local expansion strategy is designed to incorporate diverse node types into the local community, further addressing the precision limitation. We conduct numerical experiments on common community detection datasets and compare our method with classical local community detection methods. The experimental results demonstrate that the FG-based method exhibits superior performance in identifying local communities.

References

[1] B. Yang et al., "Complex network clustering algorithms," *J. Softw.*, vol. 20, no. 1, pp. 54–66, 2009.

[2] F. Radicchl et al., "Defining and identifying communities in networks," in *Proc. Nat. Acad. Sci. United States Amer.*, vol. 101, no. 9, pp. 2658–2663, 2004.

[3] M. E. Newman, "Fast algorithm for detecting community structure in networks," *Phys. Rev. E.*, vol. 69, no. 6, 2004, Art. no. 066133.

[4] M. E. Newman and M. Girvan, "Finding and evaluating community structure in networks," *Phys. Rev. E.*, 2004, vol. 69, no. 2, Art. no. 026113.

[5] C. T. Bergstrom and M. Rosvall, "Maps of random walks on complex networks reveal community structure," in *Proc. Nat. Acad. Sci.*, vol. 105, no. 4, pp. 1118–1123, 2008.

[6] Y. Xin, Z. Q. Xie, and J. Yang, "An adaptive random walk sampling method on dynamic community detection," *Expert Syst. Appl.*, vol. 58, pp. 10–19, 2016.

[7] X. Zhu, "Learning from labeled and unlabeled data with label propagation," Tech. Rep. CMU-CALD-02-107, 2002.

[8] U. N. Raghavan, R. Albert, and S. Kumara, "Near linear time algorithm to detect community structures in large-scale networks," *Phys. Rev. E Statist. Nonlinear Soft Matter Phys.*, vol. 76, no. 3Pt 2, Sep. 2007, Art. no. 036106.

[9] F. Dabaghi-Zarandi and P. Kamalipour, "Community detection in complex network based on an improved random algorithm using local and global network information," *J. Netw. Comput. Appl.*, vol. 206, 2022, Art. no. 103492.

[10] P. W. Holland, K. B. Laskey, and S. Leinhardt, "Stochastic blockmodels: First steps," *Social Netw.*, vol. 5, no. 2, pp. 109–137, 1983.

[11] J. Liu, D. Wang, S. Feng, and Y. Zhang, "Local community discovery approach based on fuzzy similarity relation," *J. Softw.*, vol. 31, pp. 3481–3491, 2020.

[12] Y. Chen and J. W. Baker, "Community detection in spatial correlation graphs: Application to non-stationary ground motion modeling," *Comput. Geosciences*, vol. 154, 2021, Art. no. 104779.

[13] L. Fanzhen et al., "Deep learning for community detection: Progress, challenges and opportunities," in *Proc. 29th Int. Joint Conf. Artif. Intell. (IJCAI-20) Surv. Track*, pp. 4981–4987, 2020.

[14] A. Clauset, "Finding local community structure in networks," *Phys. Rev. E Statist. Nonlinear Soft Matter Phys.*, vol. 72, no. 2 Pt 2, 2005, Art. no. 0 26132.

[15] W. Luo, D. Zhang, L. Ni, and N. Lu, "Multiscale local community detection in social networks," *IEEE Trans. Knowl. Data Eng.*, vol. 33, no. 3, pp. 1102–1112, Mar. 2021.

[16] P. Ronhovde and Z. Nussinov, "Multi resolution community detection for megascale networks by information-based replica correlations," *Phys. Rev. E.*, vol. 80, 2009, Art. no. 016109.

[17] Y. J. Wu et al., "Local community detection using link similarity," *J. Comput. Sci. Technol.*, vol. 27, no. 6, pp. 1261–1268, 2012.

[18] J. Chen, O. Zaïane, and R. Goebel, "Local community identification in social networks," in *Proc. Int. Conf. Adv. Social Netw. Anal. Mining*, 2009, pp. 90–94.

[19] M. E. Newman and M. Girvan, "Community structure in social and biological networks," in *Proc. Nat. Acad. Sci.*, vol. 99, no. 12, pp. 7821–7826, 2002.

[20] A. Clauset, M. E. Newman, and C. Moore, "Finding community structure in very large networks," *Phys. Rev. E.*, vol. 70, no. 6, 2004, Art. no. 066111.

[21] V. D. Blondel, "Fast unfolding of communities in large networks," *J. Statist. Mechan. Theory Experiment*, vol. 2008, 2008, Art. no. P10008.

[22] Z. Wu et al., "Improved algorithm of Louvain communities dipartition," *J. Univ. Electron. Sci. Technol. China*, vol. 42, no. 1, pp. 105–108, 2013.

[23] S. Fortunato and M. Barthélemy, "Resolution limit in community detection," in *Proc. Nat. Acad. Sci.*, vol. 104, no. 1, pp. 36–41, 2007.

[24] A. Arenas, A. Fernandez, and S. Gomez, *Analysis of the Structure of Complex Networks at Different Resolution Levels*, Bristol, U.K.: IOP Publishing, 2008.

[25] J. Xiang and K. Hu, "Limitation of multi-resolution methods in community detection," *Physica A*, vol. 391, no. 20, pp. 4995–5003, Oct. 2012.

[26] J. Reichardt and S. Bornholdt, "Statistical mechanics of community detection," *Phys. Rev. E Statal Nonlinear Soft Matter Phys.*, vol. 74, no. 2, 2006, Art. no. 016110.

[27] F. Luo, J. Z. Wang, and E. Promislow, "Exploring local community structures in large networks," *IEEE/WIC/ACM Int. Conf. on Web Intell. IEEE*, 2008, pp. 387–400.

[28] W. Luo, N. Lu, L. Ni, W. Zhu, and W. Ding, "Local community detection by the nearest nodes with greater centrality," *Inf. Sci.*, vol. 517, pp. 377–392, 2020.

[29] W. Luo, D. Zhang, H. Jiang, L. Ni, and Y. Hu, "Local community detection with the dynamic membership function," *IEEE Trans. Fuzzy Syst.*, vol. 26, no. 5, pp. 3136–3150, Oct. 2018, doi: 10.1109/TFUZZ.2018.2812148.

[30] K. Guo et al., "Local community detection algorithm based on local modularity density," *Appl. Intell.*, vol. 52, pp. 1238–1253, 2022.

[31] K. Guo et al., "A local community detection algorithm based on internal force between nodes," *Appl. Intell.*, vol. 50, pp. 328–340, Oct. 2020.

[32] B. Li, D. Kamuhanda, and K. He, "Centroid-based multiple local community detection," *IEEE Trans. Comput. Social Syst.*, vol. 11, no. 1, pp. 455–464, Feb. 2024, doi: 10.1109/TCSS.2022.3226178.

[33] A. Lancichinetti, S. Fortunato, and J. Kertész, "Detecting the overlapping and hierarchical community structure of complex networks," *New J. Phys.*, vol. 11, no. 3, 2009, Art. no. 033015.

[34] A. Lancichinetti and S. Fortunato, "Limits of modularity maximization in community detection," *Phys. Rev. E.*, vol. 84, no. 6, pp. 66122–66122, 2011.

[35] W. Luo, Z. Yan, C. Bu, and D. Zhang, "Community detection by fuzzy relations," *IEEE Trans. Emerg. Topics Comput.*, vol. 8, no. 2, pp. 478–492, Second Quarter, 2020, doi: 10.1109/TETC.2017.2751101.

[36] W. W. Zachary, "An information flow model for conflict and fission in small groups," *J. Anthropological Res.*, vol. 33, pp. 452–473, 1977.

[37] D. Lusseau, "The emergent properties of a dolphin social network," in *Proc. Biol. Sci.*, vol. 270, pp. S186–S188, 2003.

[38] V. Krebs, Political books network[EB/OL]. 2002, [Online]. Available: http://www.orgnet.com

[39] J. Yang and J. Leskovec, "Defining and evaluating network communities based on ground-truth," *Knowl. Inf. Syst.*, vol. 42, no. 1, pp. 181–213, 2012.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.

[40] A. Lancichinetti, S. Fortunato, and F. Radicchi, "Benchmark graphs for testing community detection algorithms," *Phys. Rev. E*, vol. 78, 2008, Art. no. 046110.
[41] K. Hu et al., "Significance-based multi-scale method for network community detection and its application in disease-gene prediction," *PLoS One*, vol. 15, no. 3, 2020, Art. no. e0227244.



Mengting Zhang received the BS degree from Zhejiang University, Zhejiang, China, in 2020, and the MS degree from Zhejiang University, Zhejiang,China, in 2023. Her research include research includes complex network analysis and community detection.



Weihong Bi received the PhD degree in mathematics from Zhejiang University. Currently, she is an associate professor with the Department of Mathematics, Zhejiang University. Her current research interests include scientific computing, complex network analysis, image processing, data science, and numerical algebra.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on August 24,2026 at 06:07:14 UTC from IEEE Xplore. Restrictions apply.